

[CH 8] Software

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- With the construction of stored-program computers, the next step in developing a working computer system was the software.
- Developing these programs turned out to be more difficult than expected
- Software became known as costly, slow, and bug-ridden (errors and mistakes rapidly became known as bugs)
- The term software was not coined until the late 1950s
- Programs are stored in pure binary but that is not readily readable and usable for programmers
- A more programmer-friendly notation had to be used for designing and writing software
- The initial intent was for programmers to use a friendly notation, which would then be translated into binary manually by the more lowly "coders"
- Maurice Wilkes who was the leader of Cambridge development for the EDSAC realized that the conversion could be done by the computer itself
- Wilkes handed the project over to David Wheeler who created an optimized 30-instruction program called Initial Orders which could translate programs stored on teleprinter tape and put them in computer memory
- To decrease the debugging time and cost, they decided to create a subroutine library (idea that had already been proposed by von Neumann and Goldstine)
- Many operations are common to different programs. The library would contain a number of these miniprograms that programmers could copy into their own code
- In this way, a program was one third new code and 2 thirds subroutine
- The Cambridge group published a textbook of their programming ideas titled *"The Preparation of Programs for an Electronic Computer"* in 1951 in the US to reach maximum readership
- The Cambridge book was the first programming textbook and it thus set the programming style for the early 50s. Even today the organization of subroutines still follows the model proposed in the Cambridge textbook

Flowcharts and Programming Languages

Programming consists at its very essence of 2 translations:

- The translation of a real-world problem's solution into a series of steps (algorithm) that can be performed by a machine
- The translation of this algorithm into the specific instructions used by a particular computer

Flowcharts are intended as a graphical representation of an algorithm, that served as a blueprint for programmers and an intermediate step between the analysis of the system and its implementation as a software application

FORTRAN

With larger memories, a desire for more complex programming techniques arose.

- Programs were painstakingly crafted instruction-by-instruction and took way too long
- American research labs began to experiment with automatic programming. These were systems where the programmer would write in high-level programming code which would then be translated binary instructions
- There was widespread resistance to this from programmers who did not want to change the way they were working
- Grace Murray Hopper, the first programmer for the Harvard Mark I and then the UNIVAC, went around the nation advocating for automatic programming
- She had developed a compiler called the "A-0 compiler".
 - The A-0 worked but was slow and produced inefficient programs
 - Its successor, the B-0 or "FlowMatic" was designed specifically for business applications but was also slow and inefficient
- The outstanding problem became creating an automatic programming system that would produce programs as good as those written by experienced human programmers.
 - The most successful attempt came from IBM
 - John Backus, a 29-year-old researcher at IBM, created the Formula Translator, or FORTRAN
 - FORTRAN became the first successful "programming language"
- In late 1953, Backus got approval to develop FORTRAN for the upcoming IBM Model 704
 - FORTRAN was not given much importance and some within IBM even viewed it as impossible
 - 1954 they produced a first specification, not paying much attention to the elegance of language design because everything was secondary to efficiency

- Work began on the translator in 1955 and was finished in April 1957
- The first users of FORTRAN were the Westinghouse-Bettin nuclear power facility, Maryland.
- FORTRAN was successful and in April 1958 half of the 26 Model 704's were using FORTRAN for half of their problems
- In 1959 FORTRAN II came out with improvements
- FORTRAN was the most successful and widespread scientific programming language but it was not the only one around:
 - Grace Hopper's group at UNIVAC had improved the A-0 compiler and renamed it to the Math-Matic
 - The University of Michigan created MAD
 - Carnegie-Mellon developed IT
 - The Association for Computing Machinery in the United States and its European sister organizations attempted to create an international language called the ALGOL
- Within a couple of years, FORTRAN had come to dominate
- 1961 Daniel D. McCracken published the first ever FORTRAN textbook
- Standardizing to FORTRAN meant that it was possible to exchange programs with other organizations even if they used different types of computers
- 1966 FORTRAN became the first programming language standardized by the American National Standards Institute

COBOL

- While FORTRAN was standard for scientific calculations, COBOL (Common Business Oriented Language) appeared for business applications.
- It was created in effect as a standard by the action of the US government, who did not want to have to rewrite its software every time it switched computers
- COBOL 60 came out in 1960 and was heavily influenced by Grace Murray Hopper's FlowMatic and she promoted COBOL a lot even though she was not part of its development
- Manufacturers were reluctant to adopt COBOL until the US government stated that it would not purchase any computer without a COBOL translator (unless the manufacturer could prove that the computer could not be enhanced by COBOL). Nobody attempted such a proof

Software Contractors

Programming language and other utilities helped computer users develop their software more effectively, but programming was still the most expensive part of any computer operation

- Users preferred to get ready-made programs
- Computer manufacturers began developing programs for specific industries in the late 1950s and supplying them at no additional charge
- There was no concept of software being salable at that point yet
- The first major firm in the large-systems sector was the RAND Corporation
 - It was a government owned defense contractor
 - Created software for the SAGE project, because writing the software was beyond IBM's (or anyone else's) experience
 - RAND was given the contract in 1955 because they were estimated to have the best potential of developing an adequate software-writing capability
 - To deal with this task, RAND created the Systems Development Corporation (SDC) which became America's first major software contractor
 - Out of the US's 1200 programmers, SDC employed 700 for the SAGE software project
 - The OS was 250,000 lines of code
 - The rest of the software took the total to 1,000,000 lines
- In the late 1950s and early 1960s, TRW, MITRE Corporation and Hughes Dynamics entered the large scale software contracting business
- The only others with the resources to undertake these projects were the manufacturers themselves
- After SABRE, IBM developed reservation software for Delta, Pan American, Eastern, Alitalia and BOAC
- NCR developed retailing applications
- Burroughs targeted banks
- All of them developed quite large applications
- There was a secondary market for medium sized customers. Big companies did not have the small-scale economies to target them successfully
- The first small-time software contractor was the Computer Usage Company (CUC)
 - Founded 1955 in New York by 2 scientific programmers from IBM
 - Setting up was cheap: all you needed was a programming pad and a pencil
 - Computer costs were avoided by renting time from a service bureau or using the customer's computer

- The company started with \$40,000 capital, a secretary and 4 female programmers working from one of the founders' private apartment
 - They started developing software for the Oil, Nuclear Power and similar industries
 - In the late 50s they secured NASA contracts
 - By 1959 they had 59 employees
 - 1960 the company went public and netted \$186,000
 - They used that money to buy their first computer
- By the early 1960s, there were several other start-ups:
 - Computer Science Corporation (co-founded by a participant of IBM's FORTRAN team)
 - Planning Research Corporation
 - Informatics
 - Applied Data Research
 - University Computing Company
- The first half of the 1960s saw a boom in the industry, with the number of computers increasing by an order of magnitude since the 50s and a software-hungry market with more and more big software projects
- By 1967 CUC had branched out to training, computer services, facilities management, packaged software, consultancy
 - It had 12 offices, 700 staff members and was pulling in annual sales of \$13 million
- Other than these large contractors, there were about 2800 software contracting firms in the US in 1967

Software Crisis

- By the end of the 1960s, hardware was advancing much faster in performance than software could keep up
- Existing programming technology was capable of handling 10,000 lines. Programs with 100,000 or millions of lines of code were a mess, a hassle, never-ending, unprofitable and scary for management
- The biggest debacle of the 60s was IBM's OS/360, the operating system for the System/360 line
 - It was headed by Frederick P. Brooks, who later formed the Computer Science department at the University of North Carolina
 - It was the biggest software project ever attempted, totaling at over a million lines of code
 - They attempted to fight delays and bad results by throwing more programmers at the problem, which, in hindsight, was the wrong approach
 - OS/360 was a year late and the effort of developing it caused many employees to leave
 - It was not only late, but also contained bugs that took years to fully eradicate

Software products

- December 1968 IBM unbundled. This meant software and hardware would be priced and sold separately
- Until that point, software was seen as a free good, received upon purchase of a computer
- The Mark IV was a file management system for databases, developed by Informatics
- When it was released in 1967 for the cost of \$30,000, sales were low
- After the unbundling of 1968, sales skyrocketed and Mark IV was the most successful software product for 15 years, raking in over \$100 million in total sales
- Large manufacturers like IBM remained major players, even though they weren't making the best products on the market. They could afford to lease software on a monthly basis, unlike smaller companies who needed outright sales
- Edwe